What is Ram ?

Sunday, July 21, 2024 9:52 PM

Data Structure → for Computer

[Structuring A Data {in RAM}]

↓

Random Access Memory

" different data structures stores differently in RAM"

What is RAM ?

RAM is a common computing acronym that stands for random-access memory. Sometimes it's called PC memory or just memory. In essence, RAM is your computer or laptop's short-term memory. It's where the data is stored that your computer processor needs to run your applications and open your files.

Inside your computer, RAM typically comes in the form of a rectangular flat circuit board with memory chips attached, also referred to as a memory module. Computers typically come with at least two RAM modules with room to add more, if needed. These RAM modules are critical components that work hand in hand with your computer's central processing unit (CPU) and must be working optimally for you to have a good experience.

Why is RAM important ??

If you ask your computer to perform a task, such as opening a photo to edit, and you don't have enough available RAM, your computer has to begin shuffling data around in order to open the file. Imagine someone asked you to gather up a bunch of tennis balls left on a court, but there were more than you could easily carry in your hands. You would have to start shuffling the balls around and stashing them in pockets or in your arms in order to move them all at once. Similarly, if your computer doesn't have enough RAM, it has to move data between your slower hard drive and your RAM and your processor, which causes everything to slow down.

RAM Can be 4GB, 8 GB, 12 GB, 16 GB, ...

G → Giga Byte [10^9]

B → Bytes

1 Byte is 8 bits

1 Bit is either 0 or 1

In Computers,
there Can be 16, 32, 64 bit Processor.

32-bit means that a microprocessor can execute 4 bytes of data in one instruction cycle while 64-bit means that a microprocessor can execute 8 bytes of data in one instruction cycle.

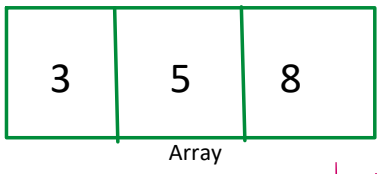
integer in 32 bit Can be stored as :

1: $\underbrace{0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ \ldots\ 0\ 0}_{31\text{ bit} \rightarrow 0}\ 1$

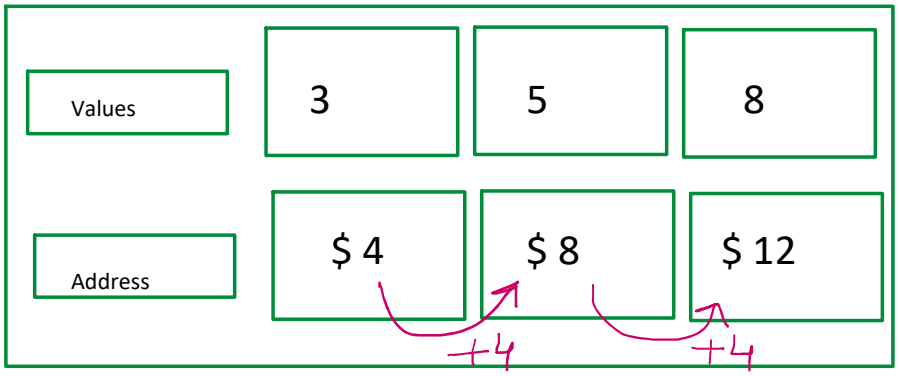
Similarly we can write in 64 bit Processors.

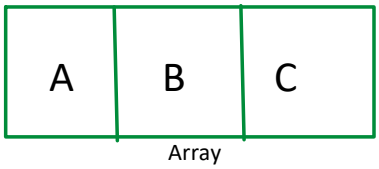
64 bit Processors >>> 32 bit Processors

ARRAY In RAM

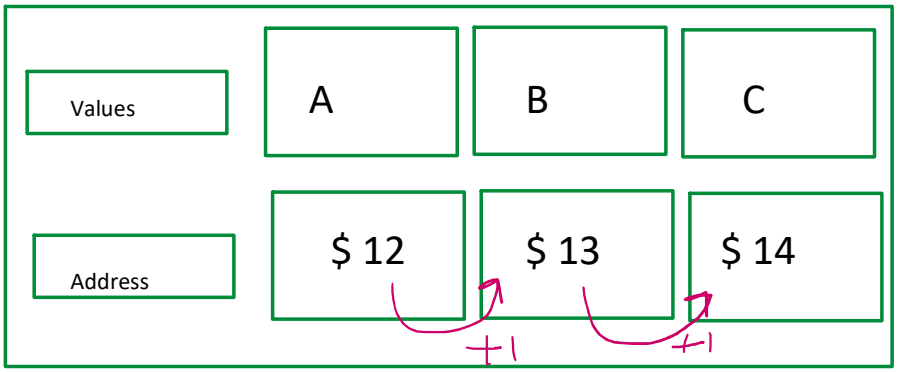


integer | float
(4 byte)

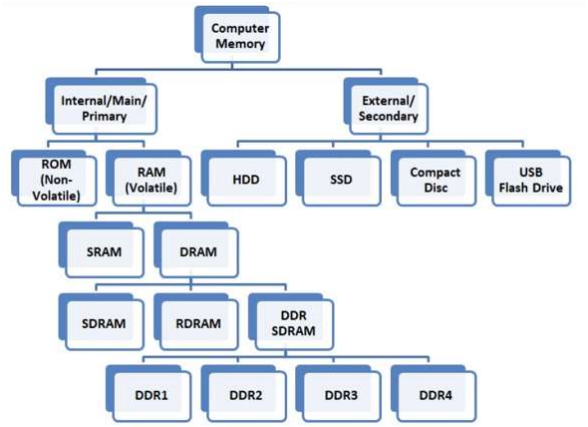


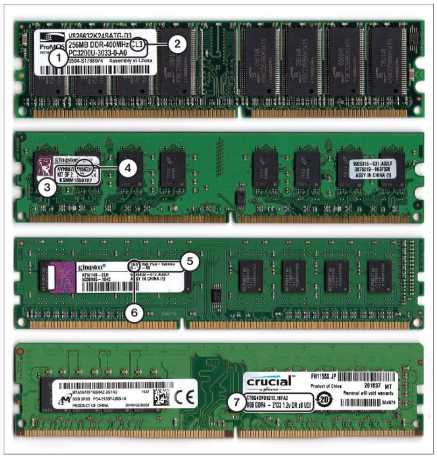


Character {ASCII}
(1 byte)



Character [illegible]
(1 byte)





1. 256MB DDR module, PC3200 (DDR400)
2. CL3 latency
3. 2GB DDR2 module (from matched set), DDR2-667 (PC2-5300)
4. CL5 latency
5. 2GB DDR3 module, PC3-10600 (DDR3-1333)
6. CL9 latency
7. 8GB DDR4 module, DDR4-2133 (PC4-17000)